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A study of the CO_2 capture pilot plant by amine absorption

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ABSTRACT

A pilot plant for CO_2 recovery of 2 ton-CO_2/day from flue gas emitted from a 500 MW coal fired power plant was tested with aqueous amine absorbents. Based on a chemical absorption/regeneration process with 30 wt% aqueous monoethanolamine (MEA) solution, we studied the CO_2 recovery as a function of flow rate and input location of absorbent, pressure and temperature of stripper, and temperature of flue gas. Experimental data include the temperature profile of the absorber/stripper and CO_2 lean/rich loading values, which remained stable during the operation. We evaluated the energy requirement for regeneration of aqueous amine and the degree of CO_2 removal. While regeneration energy of aqueous amine increased with an increase in flue gas temperature and stripper pressure, it decreased with an increase in injection height of absorbent. The optimum point in flow rate of absorbent was 1300 kg/h in 350 Sm^3/h flue gas. The regeneration energy using 30 wt% MEA solution was 3.92 GJ/ton CO_2 in 90% CO_2 removal, flue gas temperature of 40 °C, stripper pressure of 0.5 kgf/cm^2 and L/G ratio of 3.7 kg/Sm^3. In addition, corrosion rate of the pilot plant were measured in six points. It is the largest as 79.11 mpy in the bottom of the stripper tower.

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1. Introduction

The capture of CO_2 is receiving great attention from world scientists around the world. There are many technology options to separate CO_2 from flue gas; chemical and physical absorption, cryogenic separation, gas-separation membranes, bio-mineralization and vegetation [1–3]. The most common option for separating CO_2 from thermal power plants is scrubbing the gas stream using an amine solution, which can be used to dilute CO_2 streams systems such as coal combustion flue gases, which contain only 14–16% CO_2 by volume [4]. An aqueous MEA solution has been used as an industrially important absorbent because of rapid reaction, low cost of the solvent, and thermal stability. However, one of the disadvantages of amine-based absorption processes is that they require large amounts of energy in the regeneration stage of the amine absorbent. Dallos and co-workers showed that the thermal energy required for the regeneration exceeded half of the total capture energy [5]. Thus, for the development of a cost effective CO_2 capture process, many research groups have focused on the reduction of regeneration energy.

Among the alkanolamines, monoethanolamine (MEA), diethanolamine (DEA), N-methyldi-ethanolamine (MDEA), diisopropanolamine (DIPA), and 2-amino-2-methyl-1-propanol (AMP) have been widely used as chemical absorbents for removal of acid gases (CO_2 and H_2S) [6–9]. Tertiary amines such as MDEA and triisopropanolamine (TIPA) do not form carbamate due to the absence of an N–H bond. They only act as bases, contributing to the formation of bicarbonate. The advantage of tertiary amines is that the equilibrium is more easily reversed in the stripper. The use of blended amine solvents in acid gas treatment processes is receiving considerable attention from researchers [10–12]. Lee and co-workers investigated the CO_2 capture capability of 30 wt% aqueous solutions of AMP or MDEA in flue gas emitted from an LNG-based power plant [13]. Although various tests have been performed for CO_2 capture using amine absorbent, only a few experimental data on regeneration energy of CO_2 using an aqueous MEA solution have been reported in a test bed of thermal power plant (more than 1 ton CO_2/day capacity).

We studied CO_2 recovery from a newly constructed a pilot plant of which capacity is 2 ton CO_2/day. In this study, we operated the CO_2 capture process, which could treat 350 Sm^3/h flue gas, continuously for 30 days in order to analyze its long-run operation characteristics. The system used 30 wt% aqueous MEA solutions. The temperature of each tower, changing of CO_2 loadings of lean/rich solutions, and MEA concentration were analyzed. Subsequently, we have also calculated the regeneration energy of an MEA solution in the stripper using the values of degree of CO_2 removal from an on-line CO_2 gas analyzer and gas flow meters installed at the absorber. We also converted the potential heat of steam to energy in order to evaluate the energy consumption at the stripper.

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Nomenclature

G	flow rate of flue gas [Sm^3/h]
$H_{reboiler}$	regeneration energy of CO_2 [GJ/ton CO_2]
L	flow rate of MEA solution [kg/h]
R_{CO_2}	degree of CO_2 removal [−]
$V_{CO_2,in}$	CO_2 concentration in the gas entering an absorber [−]
$V_{CO_2,out}$	CO_2 concentration in the gas leaving from an absorber [−]
$V_{gas,in}$	volumetric flow rate of the flue gas entering an absorber [m^3/h]
$V_{gas,out}$	volumetric flow rate of the flue gas leaving from an absorber [m^3/h]
W_{CO_2}	mass flow rate of CO_2 captured in the absorber [kg/h]
Wt_{MEA}	weight percent of MEA in solution [−]
α_{CO_2}	CO_2 loading in mole of CO_2 per mole of MEA [mol CO_2/mol MEA]
ρ_{CO_2}	mass concentration of CO_2 in the flue gas [kg/m^3]
CR	corrosion rate in mils per year [mpy]

2. Experiment

2.1. Materials

The MEA used in the present study was obtained from BASF Chemical Co. with a mass purity of >99% and used without further purification. The aqueous solution was prepared by diluting the concentrated MEA with distilled water. The concentration of MEA was 30 wt%.

2.2. Experimental set up

The pilot plant for CO_2 recovery is shown schematically in Fig. 1. To minimize side effect of SO_x for absorbent, flue gas desulfurization system that use magnesium hydroxide ($Mg(OH)_2$) slurry was installed in first step of total processes. It managed SO_x concentration less than 20 ppm. Processes after desulfurization process are a typical amine-based absorption process consisting of an absorber (diameter = 0.4 m, total height = 23.5 m) and a stripper (diameter = 0.35 mm, total height = 17 m) with other auxiliary equipment such as a gas burner, reboiler, pump, condenser and lean/rich amine exchanger. Each tower (absorber and stripper) was packed with ring-shaped stainless steel packing material (Intalox Metal Tower Packing (IMTP), supplied by Norton Co. USA) to increase retention time and surface area for effective contact between the carbon dioxide and absorbent. The gas stream exhausted from the top of each tower was fed to an on-line gas analyzer (model ULTRAMAT 23, supplied by Siemens, France) for the analysis of CO_2 concentration in the gas stream by non-dispersion infrared method (NDIR).

The flue gas was cooled to about 40 °C in order to decrease its moisture content and subsequently introduced to the absorber. Then, CO_2 in the flue gas reacted counter-currently with a CO_2-lean MEA solution. The gases that did not react with the absorbent were emitted from the top of the absorber. At the exit line of the absorber, a water trap was installed to remove vaporized liquids from the gas stream of the absorber during the operation.

The CO_2-rich MEA solution from the bottom of the absorber was transferred to the stripper via a lean/rich plate heat exchanger in which the CO_2-rich MEA solution was heated to between 90 and 95 °C and the CO_2-lean MEA discharged from the bottom of the stripper was simultaneously cooled. In the stripper, CO_2 and MEA solutions were separated from the CO_2-rich MEA solution by heating between 110 °C and 120 °C. The mixture of steam and CO_2-rich MEA exits from the top of the stripper and is cooled in the condenser to separate the CO_2. The water vapor was sent back to the stripper from the reflux drum and 99% pure CO_2 was produced. The regenerated CO_2-lean amine solution was then cooled and recycled back to the absorber for further CO_2 removal from the flue gas. The CO_2-lean MEA solution could be fed into the absorber at different heights (H1: 16.8 m, H2: 12.6 m, H_3: 8.4 m) depending on the experimental conditions.

The compositions of flue gases from plant #8 (capacity 500 MW) are shown in Table 1. The lean/rich amine samples in the liquid phase were extracted from the absorber tower and stripper and CO_2

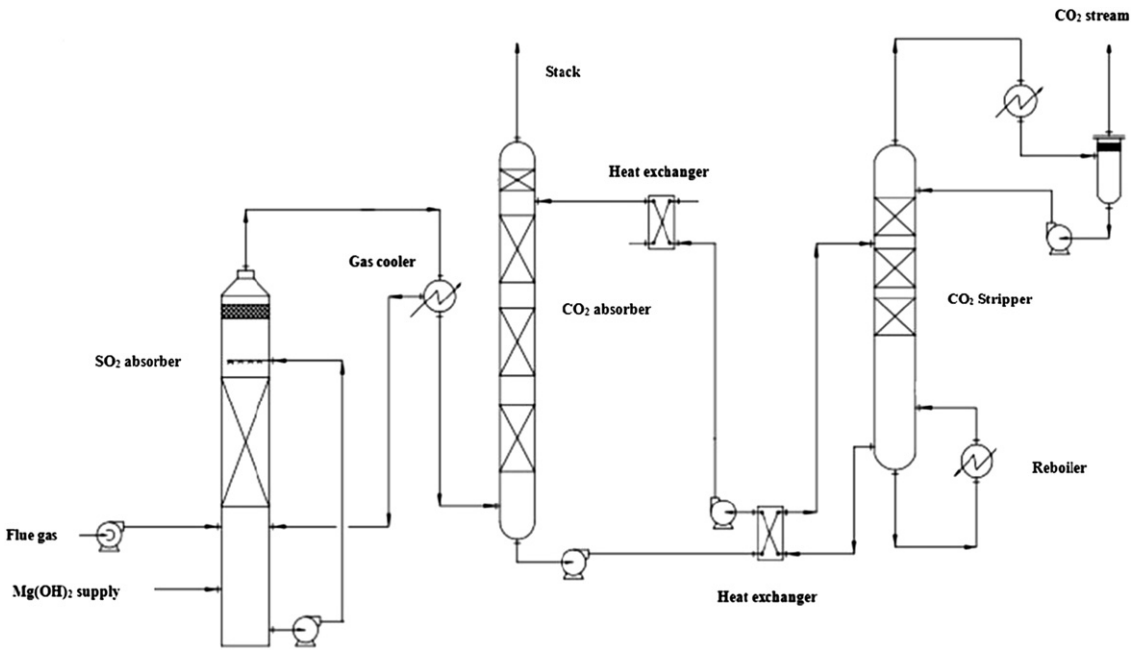


Fig. 1. Process flow diagram of 2 ton-CO_2/day pilot plant for CO_2 removal.

Table 1
Specification of flue gas emitted from 500 MW thermal power plant.

Composition	Unit	Figures
H_2O	%	2
CO_2	%	15
N_2	%	79
O_2	%	3.9
SO_2	ppm	15
Temperature	℃	53

was measured by titration method. Each sample was analyzed four times and the experimental error in the loading of CO_2 was estimated to be about ±3%. The procedure for determining absorbent content in carbon dioxide recovery facility samples assumes that all the alkalinity in the plant solutions is due to the presence of free absorbents.

2.3. Calculation of regeneration energy

The regeneration energy of CO_2($H_{reboiler}$) in the CO_2 capture process could be calculated as a ratio of the energy supplied from the reboiler ($Q_{reboiler}$) and the mass flow rate of CO_2 captured in the absorber (W_{CO_2}):

$$H_{reboiler} = Q_{reboiler}/W_{CO_2} \quad (1)$$

and W_{CO_2} can be calculated by following Eqs. (2) and (3):

$$R_{CO_2} = \left(V_{CO_2,in}V_{gas,in} - V_{CO_2,out}V_{gas,out}\right)/V_{CO_2,in}V_{gas,in} \quad (2)$$

$$W_{CO_2} = V_{gas,in}R_{CO_2}\rho_{CO_2} \quad (3)$$

Where, R_{CO_2} is the degree of CO_2 removal in the absorber, $V_{CO_2,in}$ and $V_{CO_2,out}$ denote the CO_2 concentration in the gas entering and leaving the absorber. $V_{gas,in}$ and $V_{gas,out}$ are the volumetric flow rates of the flue gas entering and leaving the absorber. Also, ρ_{CO_2} indicates the mass concentration of CO_2 in the flue gas [14].

2.4. Determination of CO_2 loadings of lean and rich MEA solution

The CO_2 loadings of lean and rich MEA solutions can be calculated based on the following equations.

$$Eq_{CO_2} = V_{NaOH} \times N_{NaOH}/V_{sample} \quad (4)$$

$$\alpha_{CO_2} = \left(Eq_{CO_2} \times MW/10\right)/Wt_{MEA} \quad (5)$$

where Eq_{CO_2} is the CO_2 content in equivalents per liter of solution, V_{NaOH} is the volume of NaOH used, N_{NaOH} is the normal concentration of NaOH, V_{sample} is the volume of sample and α_{CO_2} is the CO_2 loading in mole of CO_2 per mole of MEA. Titration procedures were performed using an automatic titrator (848 titrinoplus supplied by Metrohm USA Inc.)

2.5. Calculation of corrosion rate

The corrosion cell was cylindrical in shape and made of carbon steel 1018 with a chemical composition of 0.175% carbon, 0.75% manganese, and balance iron. A prepared specimen was mounted to the holder in the process. The corrosion cell was then connected to a potentiometer equipped with the data acquisition software. The polarization scan began when the potential of the specimen reached equilibrium [15]. Three electrochemical techniques, Tafel plot, potentiodynamic polarization, and cyclic polarization were used for corrosion measurements. Producing a Tafel plot involves polarizing a specimen in both anodic and cathodic directions in the potential range of ±1 mV from equilibrium corrosion potential (E_{corr}) and generating an anodic/cathodic polarization curve of the specimen. The obtained corrosion current density (i_{corr}) was used to calculate the corrosion rate of the specimen using the following equation:

$$CR = (0.13 \times i_{corr} \times E.W.)/(A \times D)$$

where CR is corrosion rate in mils (thousandths of an inch) per year (mpy), i_{corr} is the corrosion current density, *E.W.* is the equivalent weight of specimen, *A* is the area of specimen in cm^2, and *D* is the density of the specimen in g/cm^3 [16].

3. Results and discussion

The CO_2 recovery in the pilot plant was studied in aqueous solutions of MEA at the various absorbent flow rates, flue gas temperature, injection height of absorbent, stripper pressure, and stripper temperature.

3.1. Basic operating condition and temperature profile

The basic operating conditions for the pilot plant are given in Table 2. The flue gas was introduced to the absorber at a flow rate of 350.0 Sm^3/h and was cooled to about 40 °C. The moisture in the gas was removed before being introduced in the absorber. The temperature of CO_2-lean MEA solution fed to the absorber and reboiler of a stripper were 40 and 113 °C, respectively. The pressure of the stripper was operated at 0.50 kgf/cm^2 to prevent the boiling of MEA solution in the stripper. In our experiment, the average removal of CO_2 remained at 90% based on the guideline for the economic analysis of CO_2 capture process published by IEA Greenhouse Gas R&D Program [7].

Temperature profiles of the absorber and stripper during the operation are shown in Figs. 2 and 3. Fig. 2 is the local temperature profile of the inner part ($T_2 \sim T_5$) and the exit of the absorber (T_1) during the operation. It shows that time-averaged temperature of T_1, T_2, T_3, T_4 and T_5 is 42.3, 44.2, 47.2, 54.2, and 66.9 °C, respectively. In addition the temperature rose (about 17 °C) in the higher part of the absorber (T_5 in Fig. 2) which means that an exothermic absorption reaction takes place at this area of the tower. Emission of the highest heat of reaction occurred with the low CO_2 loading solution (lean amine) after the higher parts of the absorber (T_4 and T_5), where water evaporates from the MEA solution.

Fig. 3 shows the local temperature profile of the inner part (T_6 to T_9) of the reboiler. It shows that reboiler temperature remained almost constant during the operation (T_9_ave = 113.6 °C), though T_6 and T_7 showed some variations during the operation and the time-averaged values were T_6 (111.6 °C), T_7 (112.8 °C) and T_8 (113.8 °C).

Table 2
Operating condition.

Variable	Pilot plant	Aspen plus
Flue gas flow rate [Sm^3/h]	350.0	350.0
Solvent inventory [ton]	2.0	–
CO_2concentration in the flue gas [%]	15.2	15.2
Ratio of the flow rate of liquid and flue gas [kg/Sm^3]	3.7	4.2
CO_2-lean MEA solution feeding height [m]	12.6	12.6
Absorber temperature [℃]	40.0	40
Reboiler temperature [℃]	113.8	113.8
Stripper pressure [kgf/cm^2]	0.5	0.5

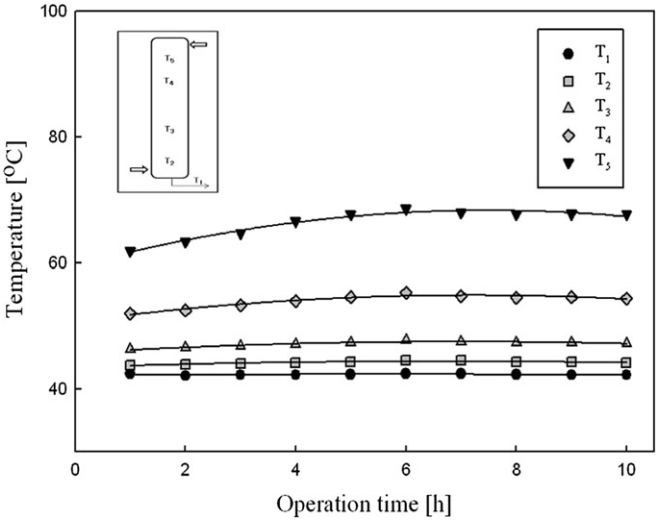


Fig. 2. Temperature profile of an absorber.

3.2. The effect of absorbent (MEA) flow rate and injection height of absorbent

The effect of absorbent flow rate on regeneration energy for different feeding positions of the MEA is shown Fig. 4. Other operation conditions are the same as basic operation conditions except ratio of the flow rate of liquid and flue gas [kg/Sm^3]. The figure shows that the optimum point for regeneration energy was 2.7–4.0 L/G ratio.

The effect of absorbent feeding height in the absorber tower above 12.6 m was small because aqueous MEA solution reacted rapidly with CO_2. Also, CO_2 loadings of rich-MEA at feeding heights above 12.6 m were almost the same as 0.50 mol CO_2/mol MEA at a 2.7 L/G ratio. Regeneration energy at a feeding height of 8.4 m increased to more than 6.0 GJ/ton CO_2 at a 2.7 L/G ratio. And CO_2 loading of rich-MEA decreased to 0.42 mol CO_2/mol MEA at a 2.7 L/G ratio.

3.3. The effect of flue gas temperature

The effect of absorbent temperature and flue gas temperature on CO_2 recovery ratio for the same concentration of MEA (30 wt%) is shown in Fig. 5. The flue gas was introduced to the absorber at a flow rate of 350.0 Sm^3/h and L/G ratio was 3.7 kg/Sm^3. The average removal of CO_2 kept 90% in an experiment. The pressure of the stripper was operated at 0.50 kgf/cm^2. Other operation conditions are the same as the basic operation conditions. The regeneration energy increased with increase in flue gas temperature from 36 to 45 °C because the absorbent absorbs more CO_2 at low temperatures. The regeneration energy is the lowest at 3.82 GJ/tons-CO_2 at 36 °C.

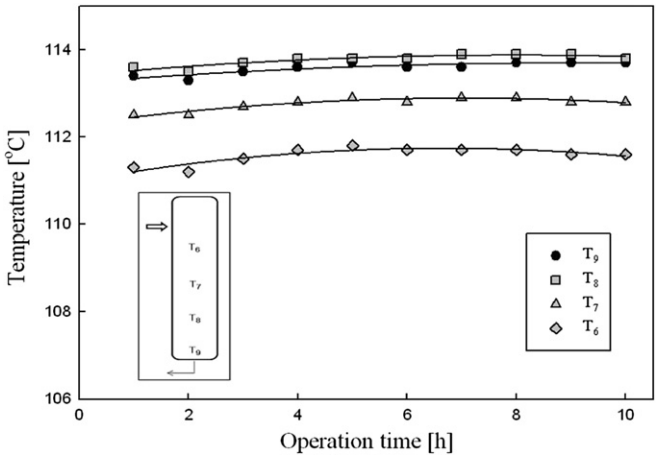


Fig. 3. Temperature profile of a stripper.

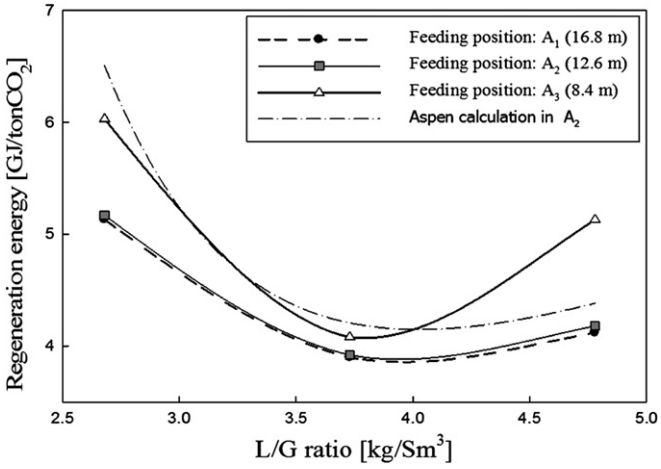


Fig. 4. The effects of absorbent flow rate and injection height.

3.4. The effect of stripper pressure and CO_2 loadings of lean and rich MEA solutions

The effect of stripper pressure on regeneration energy is shown Fig. 6. The flow rate of flue gas, input temperature of absorbents, and bottom temperature of stripper were 1300 kg/h, 40 °C, and 113 °C, respectively. The average removal of CO_2 kept 90% in an experiment. Fig. 6 shows that regeneration energy of carbon dioxide increases with an increase in stripper pressure. The regeneration energy is the lowest at 3.86 GJ/tons-CO_2 in MEA (30 wt%) at stripper pressure of about 0.35 kgf/cm^2.

The CO_2 loadings of lean and rich MEA solutions were measured every 8 h and calculated using Eqs.(4) and (5). The CO_2-loading mainly depends on the operational condition and affects both the degree of CO_2 removal and the regeneration energy. In Fig. 6, CO_2 loadings of lean and rich solutions during the operation are shown. Rich CO_2 loadings are constant as 0.50 mol CO_2/mol MEA and lean

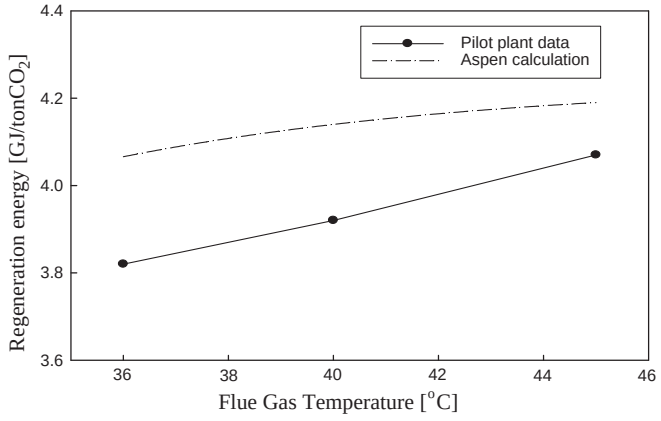


Fig. 5. Profile of flue gas temperature and regeneration energy.

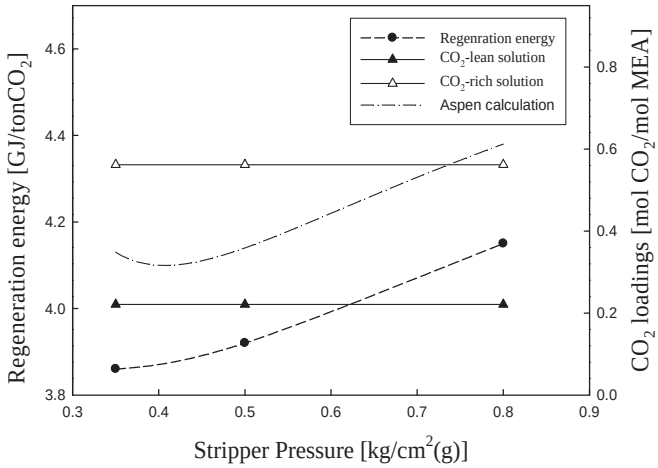


Fig. 6. The effects of stripper pressure and CO_2 loadings of lean and rich MEA solutions.

CO_2 loadings are constant as 0.15 mol CO_2/mol MEA. Thus, there is no direct relation between stripper pressure and CO_2 loadings in constant degree of CO_2 removal.

3.5. MEA concentration

The MEA concentration in the CO_2-lean/rich MEA solutions was also measured every 8 h using the titration method in basic operation conditions. The flow rate of MEA was 1300 kg/h. The regeneration energy and the degree of CO_2 removal were directly affected by the MEA concentration; the high MEA concentration decreases the reboiler duty substantially while the degree of CO_2 removal increased with an increase in MEA concentration. But the MEA of high concentrations can generally cause corrosion problems. Thus it is very important to keep the MEA concentration at a constant level for the stable operation of the process.

Fig. 7 represents the changing of the MEA concentration of CO_2-lean/rich solution during the operation. Each value increased with increasing operating time. This is mainly due to the water vapor discharge at the top of the absorber. Operating temperature at the top of the absorber is more than 60 °C. For detailed analysis, a water trap was installed at the exit of the absorber and showed that the average capture rate of liquids during the operation was 0.6 kg/day. From these results, it could be concluded that the increase of MEA concentration during the operation was mostly due to the reduction of water content in the solution. Thus, the wash water recycling zone was established at the top of the absorber to prevent evaporation of water.

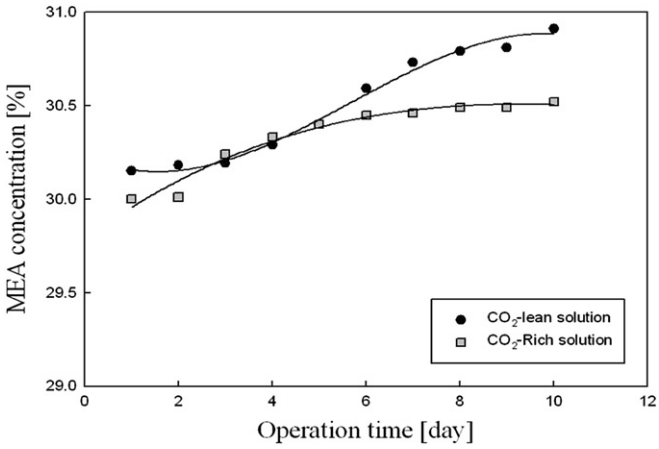


Fig. 7. Profile of lean and rich MEA concentration during the operation.

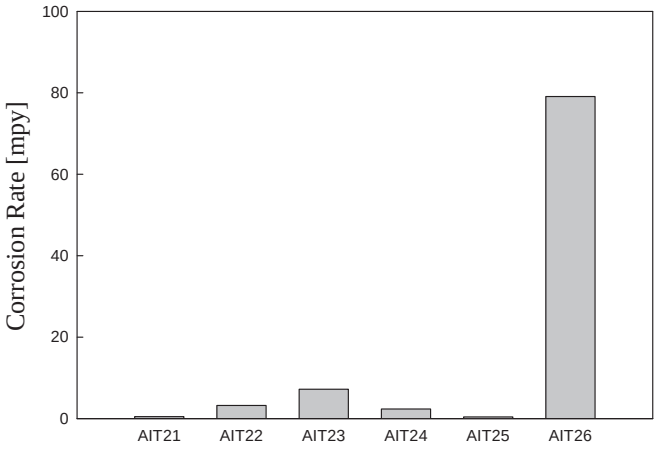


Fig. 8. Profile of corrosion rate in basic operating condition.

3.6. Corrosion rate in basic operating condition

The corrosion rate of the pilot plant was measured at six points. The three prepared specimens were located at the top (AIT 21), middle (AIT 22) and bottom (AIT 23) of the absorber tower; other three specimens were located at the top (AIT 24), middle (AIT 25) and bottom (AIT 26) of the stripper tower. Fig. 8 shows the corrosion rate is greater at the bottom of each tower. In particular, it is the largest as 79.11 mpy at the bottom of the stripper tower. Therefore, we recommend the bottom of the stripper tower be designed with corrosion-resistant materials.

4. Conclusions

We studied the recovery of CO_2 in MEA at a 2 ton-CO_2/day pilot plant as a function of temperature, concentration, and flow rate of absorbent, pressure and temperature of stripper, and flow rate and temperature of flue gas. We observed that while regeneration energy increases with an increase in flue gas temperature and stripper pressure, it decreases with an increase in injection height of absorbent. The regeneration energy had optimum point at 3.7 to 3.8 L/G ratio. MEA concentration increased slightly during the operation mainly from the reduction of water content in the solution. Thus, at wash water recycling zone needs to be installed at the top of the absorber to prevent evaporation of water. We recommend the bottom of stripper tower be designed with corrosion-resistant materials because of its high corrosion rate.

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References

[1] Kerr RA. No doubt about it, the world is warming. Science 2006;312:825.
[2] Dickens GR. Global change: hydrocarbon-driven warming. Nature 2004;429:513–5.
[3] Bharath R, Inomata H, Adschiri T, Arai K. Phase equilibrium study for the separation and fractionation of fatty oil components using supercritical carbon dioxide. Fluid Phase Equilib 1992;81:307–20.

[4] Wang R, Li DF, Liang DT. Modeling of CO_2 capture by three typical amine solutions in hollow fiber membrane contactors. Chem Eng Process 2004;43: 849–56.
[5] Dallos A, Altsach T, Kotsis L. Enthalpies of absorption and solubility of carbon dioxide in aqueous polyamine solutions. J Thermal Anal Cal 2001; 65:419–23.
[6] Bishnoi S, Rochelle GT. Physical and chemical solubility of carbon dioxide in aqueous methydiethanolamine. Fluid Phase Equilib 2004;168:241–58.
[7] Rinker EB, Ashour SS, Sandall OC. Kinetics and modeling of carbon dioxide absorption into aqueous solutions of N-methydiethanolamine. Chem Eng Sci 1995;50:755–68.
[8] Ma'mun S, Nilsen R, Svendsen HF, Juliussen O. Solubility of carbon in 30 mass% monoethanolamine and 50 mass% methydiethanolamine solutions. J Chem Eng Data 2005;50:630–4.
[9] Mandal BP, Bandyopadhyay SS. Absorption of carbon dioxide into aqueous blends of 2-amino-2-methyl-1-propanol and monoethanolamine. Chem Eng Sci 2006;61:5440–7.
[10] Mandal BP, Bandyopadhyay SS. Simultaneous absorption of carbon dioxide and hydrogen sulfide into aqueous blends of 2-amino-2-methyl-1-propanol and diethanolamine. Chem Eng Sci 2005;60:6438–51.
[11] Bosch H, Versteeg GF, Van Swaaij WPM. Kinetics of the reaction of CO_2 with the sterically hindered amine 2-amino-2-methylpropanol at 298 K. Chem Eng Sci 1990;45:1167–73.
[12] Chakraborty AK, Astarita G, Bischoff KB. CO_2 absorption in aqueous solutions of hindered amines. Chem Eng Sci 1986;41:997–1003.
[13] Lee S, Maken S, Park JW, Song HJ, Park JJ, Shim JG, et al. A study on the carbon dioxide recovery from 2 ton-CO_2/day pilot plant at LNG based power plant. Fuel 2008;87:1734–9.
[14] Lee JH, Kim JH, Lee IY, Jang KR, Shim JG. Experimental study of bench scale CO_2 capture unit by amine absorption. Sci China Technol Sci 2011:53.
[15] ASTM standard G1-90 (Re-approved 1999), Standard practice for preparing, cleaning and evaluating corrosion test specimens; 1999.
[16] Veawab A, Nainar M. Corrosion in CO_2 capture unit using MEA-piperazine blends. Energy Procedia 2009;1:231–5.